

ON THE LEAST QUADRATIC NON-RESIDUE

YUK-KAM LAU AND JIE WU

ABSTRACT. We prove that for almost all real primitive characters χ_d of modulus $|d|$, the least positive integer n_{χ_d} at which χ_d takes a value not equal to 0 and 1 satisfies $n_{\chi_d} \ll \log |d|$, and give a quite precise estimate on the size of the exceptional set.

1. INTRODUCTION

Let $q \geq 2$ be an integer and χ a non principal Dirichlet character modulo q . Here the evaluation of the least integer n_χ among all positive integers n for which $\chi(n) \neq 0, 1$ is referred as Linnik's problem. In case χ coincides with the Legendre symbol, n_χ is a least quadratic non-residue. Concerning the size of n_χ , Pólya-Vinogradov's inequality

$$(1.1) \quad \max_{x \geq 1} \left| \sum_{n \leq x} \chi(n) \right| \ll q^{1/2} \log q$$

implies trivially $n_\chi \ll q^{1/2} \log q$. But for prime q , Vinogradov [21] proved the better bound

$$(1.2) \quad n_\chi \ll q^{1/(2\sqrt{e})} (\log q)^2$$

by combining a simple argument with (1.1). He also conjectured that $n_\chi \ll_\varepsilon q^\varepsilon$ for all integers $q \geq 2$ and any $\varepsilon > 0$. Under the Generalized Riemann Hypothesis (GRH), Linnik [16] settled this conjecture, and later Ankeny [1] gave a sharper estimate

$$(1.3) \quad n_\chi \ll (\log q)^2$$

(still assuming GRH). Burgess ([3], [4], [5]) wrote a series of important papers on sharpening (1.1). His well known estimate on character sums is as follows: For any $\varepsilon > 0$, there is $\delta(\varepsilon) > 0$ such that

$$(1.4) \quad \left| \sum_{n \leq x} \chi(n) \right| \ll_\varepsilon x q^{-\delta(\varepsilon)}$$

provided $x \geq q^{1/3+\varepsilon}$. The last condition can be improved to $x \geq q^{1/4+\varepsilon}$ if q is cubefree. When q is prime, Burgess deduced, via (1.4) and Vinogradov's argument,

$$(1.5) \quad n_\chi \ll_\varepsilon q^{1/(4\sqrt{e})+\varepsilon}.$$

An entirely straightforward modification yields

$$n_\chi \ll_\varepsilon \begin{cases} q^{1/(4\sqrt{e})+\varepsilon} & \text{if } q \text{ is cubefree,} \\ q^{1/(3\sqrt{e})+\varepsilon} & \text{otherwise.} \end{cases}$$

(However we would remark that we could not find this result in literature.)

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Let us now focus on real primitive characters. Denote $\mathcal{D}$ (resp. $\mathcal{D}(Q)$) to be the set of fundamental discriminants d (resp. with $|d| \leq Q$), that is, the set of non-zero integers d which are products of coprime factors of the form $-4, 8, -8, p'$ where $p' := (-1)^{(p-1)/2}p$ (p odd prime). Also, we write $\mathcal{K}$ (resp. $\mathcal{K}(Q)$) for the set of real primitive characters (resp. with modulus $q \leq Q$). Then there is a bijection between $\mathcal{D}$ and $\mathcal{K}$ given by

$$d \mapsto \chi_d(\cdot) = \left(\frac{d}{\cdot}\right)_K$$

where $\left(\frac{d}{\cdot}\right)_K$ is the Kronecker symbol. Note that the modulus of χ_d equals $|d|$ and

$$(1.6) \quad |\mathcal{D}(Q)| = |\mathcal{K}(Q)| = \frac{6}{\pi^2}Q + O(Q^{1/2}).$$

In the opposite direction of (1.2), Fridlender [12], Salié [20] and Chowla & Turán (see [10]) independently showed that there are infinitely many primes p for which

$$(1.7) \quad n_{\chi_{p'}} \gg \log p,$$

or in other words, $n_{\chi_{p'}} = \Omega(\log p)$. Under GRH, Montgomery [18] gave a stronger result $n_{\chi_{p'}} = \Omega(\log p \log_2 p)$, where $\log_k$ denotes the k -fold iterated logarithm. Without any assumption Graham & Ringrose [13] obtained $n_{\chi_{p'}} = \Omega(\log p \log_3 p)$. In view of these results, it is natural to wonder what is the size of the majority of $n_{\chi_{p'}}$, or more generally n_{χ_d} . Indeed the density of p' for which $n_{\chi_{p'}}$ satisfies (1.7) is low. This can be seen from Erdős' result [11],

$$(1.8) \quad \lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} n_{\chi_{p'}} = \text{constant},$$

where $\pi(x)$ denotes the number of primes up to x . This result is extended and refined by Elliott in [7] and [8]. Using (1.8) or its refinement in [7], it follows, for any fixed constant $\delta > 0$, that

$$(1.9) \quad \sum_{\substack{p \leq x \\ n_{\chi_{p'}} \geq \delta \log p}} 1 \ll_{\delta} \frac{x}{(\log x)^2}.$$

In [6], Duke & Kowalski indicated: Let $\alpha > 1$ be given. Denote by $N(Q, \alpha)$ the number of primitive characters χ (not necessarily real) of modulus $q \leq Q$ such that $\chi(n) = 1$ for all $n \leq (\log Q)^{\alpha}$ and $(n, q) = 1$. Then one has

$$N(Q, \alpha) \ll_{\varepsilon} Q^{2/\alpha+\varepsilon}$$

for all $\varepsilon > 0$. Therefore

$$|\{d : |d| \leq Q : n_{\chi_d} \geq (\log Q)^{\alpha}\}| \ll_{\varepsilon} Q^{2/\alpha+\varepsilon}.$$

However, in view of (1.6) this result is non-trivial only when $\alpha > 2$ and it tells that $n_{\chi_d} \leq (\log |d|)^{2+\varepsilon}$ for almost all fundamental discriminants d . Very recently Baier [2] improved $2 + \varepsilon$ to $1 + \varepsilon$ by using the large sieve inequality of Heath-Brown [14] for real primitive characters. However, the argument is unable to cover the case $\alpha = 1$ or to provide information on the sparsity of the primes p with $n_{\chi_{p'}} \gg \log p$ as in (1.9).

The purpose of this paper is two-fold. The first goal is to supplement the case $\alpha = 1$, using the large sieve inequality of Elliott-Montgomery-Vaughan (see [9] and [19]). We obtain an almost all result, which is strong enough to yield a tighter estimate on the low density of exceptional non-residues than in (1.9).

Theorem 1.1. *For $2 \leq P \leq Q$, define*

$$\mathcal{E}(Q, P) := \{d \in \mathcal{D}(Q) : \chi_d(p) = 1 \text{ for } P < p \leq 2P \text{ and } p \nmid |d|\}.$$

Then there are two absolute positive constants C and c such that

$$(1.10) \quad |\mathcal{E}(Q, P)| \ll Q e^{-c(\log Q)/\log_2 Q}$$

holds uniformly for $Q \geq 10$ and $C \log Q \leq P \leq (\log Q)^2$. In particular we have

$$(1.11) \quad n_{\chi_d} \ll \log |d|$$

for all but except $O(Q e^{-c(\log Q)/\log_2 Q})$ characters $\chi_d \in \mathcal{K}(Q)$.

Sections 2 and 3 are devoted to the proof of Theorem 1.1.

Our second goal is to point out that the upper bound (1.10) for exceptional real primitive characters set is quite optimal.

Theorem 1.2. *For any fixed constant $\delta > 0$, there are a sequence of positive real numbers $\{Q_n\}_{n=1}^\infty$ with $Q_n \rightarrow \infty$ and a positive constant c such that*

$$(1.12) \quad \sum_{\substack{Q_n^{1/2} < p \leq Q_n \\ n_{\chi_p} \geq \delta \log p}} 1 \gg_\delta Q_n e^{-c(\log Q_n)/\log_2 Q_n}.$$

Further if we assume that both $\mathbf{L}_1(s, P_y)$ and $\mathbf{L}_4(s, P_y)$ defined in (4.3) below have no exceptional zeros in the region (4.4), then (1.12) holds for all $Q \geq 10$.

Theorem 1.2 is essentially due to Graham & Ringrose [13]. Graham & Ringrose considered a problem of the quasi-random graphs (Paley graphs) which leads to study the lower bound for the sum of the right-hand side of (5.5) below. Their work will be the core of our proof. In Sections 4 and 5, we shall provide the salient points along the line of arguments in [13] to prove Theorem 1.2.

2. A LARGE SIEVE INEQUALITY OF MONTGOMERY-VAUGHAN

Our key tool for proving Theorem 1.1 is a large sieve inequality of Montgomery & Vaughan in ([19], page 1050) following from ([19], Lemma 2). Here we state a slightly refined version. Their original statement absorbs the factors $(6/\log P)^j$ and $\{6/(\log P)^2\}^j$ in the implied constant. We reproduce here their proof with a minuscule modification.

Lemma 2.1. *We have*

$$(2.1) \quad \sum_{d \in \mathcal{D}(Q)} \left| \sum_{P < p \leq 2P} \frac{\chi_d(p)}{p} \right|^{2j} \ll Q \left(\frac{6j}{P \log P} \right)^j + \left(\frac{6P}{(\log P)^2} \right)^j$$

uniformly for $2 \leq P \leq Q$ and $j \geq 1$. The implied constant is absolute.

Proof. Since $\chi_d(n)$ is completely multiplicative on n , we can write

$$\left(\sum_{P < p \leq 2P} \frac{\chi_d(p)}{p} \right)^j = \sum_{P^j < m \leq (2P)^j} \frac{a_j(m)}{m} \chi_d(m),$$

where

$$a_j(m) := |\{(p_1, \dots, p_j) : p_1 \cdots p_j = m, P < p_i \leq 2P\}|.$$

By Lemma 2 of [19] with the choice of parameters

$$X = P^j, \quad Y = (2P)^j \quad \text{and} \quad a_m = a_j(m)/m,$$

it follows that as $a_j(m_1)a_j(m_2) \leq a_{2j}(n^2)$ for $n^2 = m_1m_2$,

$$(2.2) \quad \sum_{d \in \mathcal{D}(Q)} \left| \sum_{P < p \leq 2P} \frac{\chi_d(p)}{p} \right|^{2j} \ll Q \sum_{P^j < n \leq (2P)^j} \frac{a_{2j}(n^2)}{n^2} + \left(\sum_{P < p \leq 2P} \frac{1}{p^{1/2}} \right)^{2j}.$$

Writing $n = p_1^{\nu_1} \cdots p_i^{\nu_i}$ with $\nu_1 + \cdots + \nu_i = j$, we have

$$\begin{aligned} a_{2j}(n^2) &= \frac{(2j)!}{(2\nu_1)! \cdots (2\nu_i)!} \\ &= \frac{(2j)!}{j!} \frac{\nu_1!}{(2\nu_1)!} \cdots \frac{\nu_i!}{(2\nu_i)!} a_j(n). \end{aligned}$$

From this, it is easy to see $a_{2j}(n^2) \leq j^j a_j(n)$, and thus

$$\begin{aligned} \sum_{P^j < n \leq (2P)^j} \frac{a_{2j}(n^2)}{n^2} &\leq j^j \sum_{P^j < n \leq (2P)^j} \frac{a_j(n)}{n^2} \\ &= \left(j \sum_{P < p \leq 2P} \frac{1}{p^2} \right)^j \\ &\leq \left(\frac{6j}{P \log P} \right)^j. \end{aligned}$$

Inserting this into (2.2) and using the estimate

$$\sum_{P < p \leq 2P} \frac{1}{p^{1/2}} \leq \frac{6P^{1/2}}{\log P},$$

we obtain the required result (2.1). $\square$

3. PROOF OF THEOREM 1.1

Define

$$\mathcal{E}^*(Q, P) := \{d \in \mathcal{D}(Q) : Q^{1/2} \leq |d| \leq Q \text{ and } \chi_d(p) = 1 \ (P < p \leq 2P, p \nmid |d|)\}.$$

Let $C \log Q \leq P \leq (\log Q)^2$. For $d \in \mathcal{E}^*(Q, P)$, we invoke the prime number theorem to deduce

$$\begin{aligned} \sum_{P < p \leq 2P} \frac{\chi_d(p)}{p} &= \sum_{P < p \leq 2P} \frac{1}{p} - \sum_{P < p \leq 2P, p \parallel |d|} \frac{1}{p} \\ &\geq \frac{\log 2 + o(1)}{\log P} - \frac{\{1 + o(1)\} \log Q}{P \log_2 Q} \\ &\geq \frac{\log 2 - 2/C + o(1)}{\log P} \\ &> \frac{1}{2 \log P}, \end{aligned}$$

provided C is sufficiently large. It is apparent from (2.1) that

$$\begin{aligned} \frac{|\mathcal{E}^*(Q, P)|}{(2 \log P)^{2j}} &\leq \sum_{d \in \mathcal{D}(Q)} \left| \sum_{P < p \leq 2P} \frac{\chi_d(p)}{p} \right|^{2j} \\ &\ll Q \left(\frac{6j}{P \log P} \right)^j + \left(\frac{6P}{(\log P)^2} \right)^j. \end{aligned}$$

Hence we obtain

$$|\mathcal{E}^*(Q, P)| \ll Q \left(\frac{12j \log P}{P} \right)^j + (12P)^j$$

uniformly for $C \log Q \leq P \leq (\log Q)^2$ and $j \geq 1$. Taking

$$j = \left\lfloor \frac{\log Q}{48 \log P} \right\rfloor + 1,$$

a simple calculation shows that

$$|\mathcal{E}^*(Q, P)| \ll Q e^{-c(\log Q)/\log_2 Q}$$

with $c = (\log 2)/48$. This implies (1.10).

Finally let

$$\mathcal{E}^*(Q) := \{d \in \mathcal{D}(Q) : d \leq Q^{1/2}\} \cup \mathcal{E}^*(Q, C \log Q).$$

Then by (1.10), we have

$$|\mathcal{E}^*(Q)| \ll Q e^{-c(\log Q)/\log_2 Q},$$

and for any $d \in \mathcal{D}(Q) \setminus \mathcal{E}^*(Q)$ there is a prime number $p \asymp \log Q \asymp \log |d|$ such that $\chi_d(p) \neq 1$, which implies (1.11). The proof is complete.

4. GRAHAM-RINGROSE'S METHOD

In this section, we shall state and extend the main results of ([13], Theorems 2, 3 and 4) for our purposes. For characters of certain moduli, Graham & Ringrose [13] obtained a wide zero-free region and good zero density estimates for the corresponding Dirichlet L -functions. The main ingredient of their method is an q -analogue of van der Corput's result, which can be stated as follows: *Suppose that $q = 2^\nu r$, where $0 \leq \nu \leq 3$ and r is an odd squarefree integer, and that χ is a non-principal character mod q . Let p be the largest prime factor of q . Suppose that k is a non-negative integer, and $K = 2^k$. Finally, assume that $N \leq M$. Then*

$$(4.1) \quad \sum_{M < n \leq M+N} \chi(n) \ll M^{1 - \frac{k+3}{8K-2}} p^{\frac{k^2+3k+4}{32K-8}} q^{\frac{1}{8K-2}} d(q)^{\frac{32k^2+11k+8}{16K-4}} (\log q)^{\frac{k+3}{8K-2}} \sigma_{-1}(q),$$

where $\sigma_a(q) := \sum_{d|q} d^a$ and $d(q) := \sigma_0(q)$. The implied constant is absolute.

Recall that for any odd prime p ,

$$\chi_8(p) = \left(\frac{2}{p} \right), \quad \chi_{q'}(p) = \left(\frac{q}{p} \right)_K = \left(\frac{q}{p} \right) \quad (q \text{ odd prime, } q' := (-1)^{(q-1)/2} q)$$

by definition. For squarefree $m \geq 2$, the character $\chi_m := \prod_{p|m} \chi_{p'}$ for odd m or $\chi_m := \chi_8 \chi_{m'}$ for $m = 2m'$ is a real primitive of modulus m or $4m$, respectively. By convention, we set $\chi_1 \equiv 1$. Moreover, if χ_4 is the real primitive character mod 4, i.e. $\chi_4(n) = (-1)^{(n-1)/2}$ for odd n , then $\chi_{4m} := \chi_4 \chi_m$ is also a real primitive character of modulus $4m$.

Let

$$(4.2) \quad P_y := \prod_{p \leq y} p = e^{\{1+o(1)\}y} \quad (y \rightarrow \infty),$$

and define for $\ell = 1$ or 4,

$$(4.3) \quad \mathbf{L}_\ell(s, P_y) := \prod_{m|P_y} L(s, \chi_{\ell m}),$$

where $L(s, \chi_{\ell m})$ is the Dirichlet L -function associated to $\chi_{\ell m}$. Denote by $N_\ell(\alpha)$ the number of zeros of $\mathbf{L}_\ell(s, P_y)$ in the rectangle

$$\alpha \leq \sigma \leq 1 \quad \text{and} \quad |\tau| \leq \log P_y.$$

Here and in the sequel we implicitly define the real numbers σ and τ by the relation $s = \sigma + i\tau$.

The next lemmas 4.1, 4.2 and 4.3 are trivial extensions of Theorems 2, 3 and 4 of [13], respectively.

Lemma 4.1. *Let $y \geq 100$. Then there is an absolute positive constant C_1 such that the L -function $\prod_{\ell=1,4} \mathbf{L}_\ell(s, P_y)$ has at most one zero in the region*

$$(4.4) \quad \sigma \geq 1 - \frac{C_1(\log_2 P_y)^{1/2}}{\log P_y} \quad \text{and} \quad |\tau| \leq \log P_y.$$

The exceptional zero, if exists, is real.

Proof. As the crucial estimate (4.1) holds for all non-principal primitive characters of modulus $q = 2^\nu r \geq 2$ with $0 \leq \nu \leq 3$ and r being odd squarefree. Consider the case $\nu = 0$ or 3 , and $\nu = 2$ or 3 , respectively. We see that (4.1) applies to χ_m and χ_{4m} for any $m|P_y$. It follows that ([13], Lemma 6.1) is valid for these characters. Proceeding with the same argument, we have ([13], Lemma 6.2) for our L -function $\prod_{\ell=1,4} \mathbf{L}_\ell(s, P_y)$ in place of $\mathbf{L}(s, P_y)$ there. Then the same proof of ([13], Theorem 2) will give the desired result. (Note that the value of ϕ suffers a negligible change when P_y is replaced by $4P_y$ or $8P_y$.) The exceptional zero must be real, for otherwise, its conjugate is another zero in the specified region. $\square$

Lemma 4.2. *Let C_1 be as in Lemma 4.1. There is a sequence of positive real numbers $\{y_n\}_{n=1}^\infty$ with $y_n \rightarrow \infty$ such that both $\mathbf{L}_1(s, P_{y_n})$ and $\mathbf{L}_4(s, P_{y_n})$ have no zeros in the region*

$$(4.5) \quad \sigma \geq 1 - \eta(y_n) \quad \text{and} \quad |\tau| \leq \log P_{y_n},$$

where

$$\eta(y) := \frac{C_1(\log_2 P_y)^{1/2}}{2 \log P_y}.$$

Proof. Similar to ([13], Theorem 3), our proof is also based on an interesting argument attributed to Maier [17]. Suppose that for some y , the product $\mathbf{L}_1(s, P_y) \mathbf{L}_4(s, P_y)$ has an exceptional zero in the region (4.4). That is, it has a real zero $\beta > 1 - 2\eta(y)$. In view of (4.2), we can take $y_n \geq y$ such that

$$(4.6) \quad \eta(y_n) < 1 - \beta < 2\eta(y_n).$$

By Lemma 4.1, β is the only exceptional zero of $\prod_{\ell=1,4} \mathbf{L}_\ell(s, P_{y_n})$ in the region

$$\sigma > 1 - 2\eta(y_n) \quad \text{and} \quad |\tau| \leq \log P_{y_n}.$$

Together with the first inequality in (4.6), this forces $\prod_{\ell=1,4} \mathbf{L}_\ell(s, P_{y_n})$ to have no zero in the region (4.5). It follows that we can find a sequence of positive real numbers $\{y_n\}_{n=1}^\infty$ with $y_n \rightarrow \infty$ such that both $\mathbf{L}_1(s, P_{y_n})$ and $\mathbf{L}_4(s, P_{y_n})$ have no zero in this region. $\square$

Lemma 4.3. *Let $\ell = 1$ or 4 and $y \geq 100$. Then there is an absolute constant C_2 such that*

$$(4.7) \quad N_\ell(\alpha) \ll \begin{cases} \exp \left\{ \frac{C_2(1-\alpha) \log P_y}{\sqrt{\log_2 P_y}} + \frac{\log_3 P_y}{2} \right\} & \text{if } \alpha \geq 1 - \eta_1(y), \\ \exp \left\{ \frac{C_2(1-\alpha) \log P_y}{\log(1/(1-\alpha))} \right\} & \text{if } \alpha < 1 - \eta_1(y), \end{cases}$$

where

$$k_0(y) := [(\log_2 P_y)^{1/2}] \quad \text{and} \quad \eta_1(y) := \frac{k_0(y)}{2(2^{k_0(y)} - 2)}.$$

Proof. The case of $\ell = 1$ has been done in ([13], Sections 7 and 8) and $N_4(\alpha)$ can be treated in the same way by applying (4.1) to our χ_{4m} . $\square$

5. PROOF OF THEOREM 1.2

In this section, we denote by p and q prime numbers. Define

$$\mathbb{P}_y := \{p : p \equiv 1 \pmod{4} \text{ and } \chi_p(q) = 1 \text{ for all } q \leq y\}.$$

Clearly we have $n_{\chi_p} > y$ for any $p \in \mathbb{P}_y$. We shall first show that the set $\mathbb{P}_y$ is not too small for suitable y .

Proposition 5.1. *Let $\delta > 0$ be a fixed small constant and $y(x)$ be an strictly increasing function defined on $[120, \infty)$ satisfying*

$$(5.1) \quad (\log x) e^{-\delta(\log_2 x)^{1/2}} \leq y(x) \leq \delta(\log x) \log_3 x.$$

Then there is a positive constant $c = c(\delta)$ and a sequence of positive real numbers $\{x_n\}_{n=1}^\infty$ with $x_n \rightarrow \infty$ such that

$$(5.2) \quad \sum_{\substack{x_n^{1/2} < p \leq x_n \log x_n \\ p \in \mathbb{P}_{y(x_n)}}} 1 \gg x_n e^{-cy(x_n)/\log y(x_n)}.$$

Further if we assume that both $\mathbf{L}_1(s, P_y)$ and $\mathbf{L}_4(s, P_y)$ have no zeros in the region (4.4) for all $y \geq 100$, then there is a positive constant c such that for all $x \geq 100$ we have

$$(5.3) \quad \sum_{\substack{x^{1/2} < p \leq x \log x \\ p \in \mathbb{P}_y}} 1 \gg x e^{-cy(x)/\log y(x)}.$$

Proof. First let $10 \leq y \leq x^{1/2}$. As usual, $\pi(y)$ denotes the number of prime numbers $\leq y$. Clearly we have

$$(5.4) \quad 2^{-\pi(y)-1} (1 + \chi_4(p)) \prod_{q \leq y} (1 + \chi_p(q)) = \begin{cases} 1 & \text{if } p \in \mathbb{P}_y, \\ 0 & \text{if } p \notin \mathbb{P}_y. \end{cases}$$

When p and q are odd primes with $p \equiv 1 \pmod{4}$, i.e. $\chi_4(p) = 1$, we infer by quadratic reciprocity law that

$$\chi_p(q) = \left(\frac{p}{q}\right) = \left(\frac{q}{p}\right) = \chi_{q'}(p) \quad (q' := (-1)^{(q-1)/2} q).$$

Note also for odd prime p ,

$$\chi_p(2) = \left(\frac{p}{2}\right)_K = \left(\frac{2}{p}\right) = \chi_8(p).$$

Thus we can replace $\chi_p(q)$ by (qp) in (5.4) to write

$$\sum_{\substack{x^{1/2} < p \leq x \log x \\ p \in \mathbb{P}_y}} 1 = \frac{1}{2^{\pi(y)+1}} \sum_{x^{1/2} < p \leq x \log x} (1 + \chi_4(p)) \prod_{q \leq y} \left(1 + \left(\frac{q}{p}\right)\right).$$

It is convenient to introduce the weight factor $(\log p) (e^{-p/(2x)} - e^{-p/x})$ to the summands,

$$\begin{aligned} \sum_{\substack{x^{1/2} < p \leq x \log x \\ p \in \mathbb{P}_y}} 1 &\geq \frac{1}{2^{\pi(y)+2} \log x} \sum_{x^{1/2} < p \leq x \log x} (\log p) (e^{-p/(2x)} - e^{-p/x}) \times \\ &\quad \times (1 + \chi_4(p)) \prod_{q \leq y} \left(1 + \left(\frac{q}{p}\right)\right). \end{aligned}$$

We want to relax the range of the sum over p . To this end, we observe that by the prime number theorem and integration by parts,

$$\begin{aligned} \frac{1}{2^{\pi(y)} \log x} \sum_{x \log x < p \leq x^2} (\log p) (e^{-p/(2x)} - e^{-p/x}) (1 + \chi_4(p)) \prod_{q \leq y} \left(1 + \left(\frac{q}{p}\right)\right) \\ \ll \sum_{x \log x < p \leq x^2} (e^{-p/(2x)} - e^{-p/x}) \\ \ll x^{1/2} / \log x. \end{aligned}$$

Combining this with the preceding inequality, we obtain

$$(5.5) \quad \sum_{\substack{x^{1/2} < p \leq x \log x \\ p \in \mathbb{P}_y}} 1 \geq \frac{1}{(\log x) 2^{\pi(y)+2}} \sum_{m|P_y} (S_x(m) + S_x(4m)) + O\left(\frac{x^{1/2}}{\log x}\right),$$

where $\ell = 1$ or 4 , and

$$S_x(\ell m) := \sum_{x^{1/2} < p \leq x^2} (\log p) (e^{-p/(2x)} - e^{-p/x}) \chi_{\ell m}(p).$$

By the Perron formula, we can write

$$(5.6) \quad S_x(\ell m) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} -\frac{L'}{L}(s, \chi_{\ell m}) (2^s - 1) \Gamma(s) x^s ds + O(x^{1/2} \log x).$$

We shift the line of integration to $\sigma = -34$. The function $(2^s - 1) \Gamma(s) x^s$ has no pole in the strip $-34 \leq \sigma \leq 2$ since the pole of $\Gamma(s)$ at $s = 0$ is canceled by the zero of $(2^s - 1)$. Thus the only poles of the integrand in (5.6) occur at $s = 1$ if $\ell m = 1$ (note that $L(s, \chi_1)$ is the Riemann ζ -function), or at the zeros $\rho(\ell m) = \beta(\ell m) + i\gamma(\ell m)$ of $L(s, \chi_{\ell m})$. It follows that

$$S_x(\ell m) = \delta_{\ell m, 1} x - \sum_{\rho(\ell m)} (2^{\rho(\ell m)} - 1) \Gamma(\rho(\ell m)) x^{\rho(\ell m)} + O(x^{1/2} \log x),$$

where $\delta_{j,1} = 1$ if $j = 1$ and 0 otherwise, and the sum is over all zeros with $0 \leq \beta(\ell m) < 1$.

We write $N(T, \chi_{\ell m})$ for the number of zeros of $L(s, \chi_{\ell m})$ in the rectangle $0 < \beta(\ell m) < 1$ and $|\gamma| \leq T$. Then we have the classical bound

$$(5.7) \quad N(T, \chi_{\ell m}) \ll T \log(Tm),$$

which implies, for any $\alpha \in (0, 1)$,

$$(5.8) \quad N_\ell(\alpha) \leq \sum_{m|P_y} N(\log P_y, \chi_{\ell m}) \ll 2^{\pi(y)} y^2.$$

On the other hand, by means of $(2^s - 1) \Gamma(s) x^s \ll x^\sigma |\tau| e^{-(\pi/2)|\tau|}$, the contribution of the zeros with $|\gamma(\ell m)| \geq \log P_y$ to $S_x(\ell m)$ is $\ll 1$. Let ε be an arbitrarily small positive number. The zeros with $\beta(\ell m) \leq 1 - \varepsilon$ and $|\gamma(\ell m)| \leq \log P_y$ contribute

$$\ll x^{1-\varepsilon} N(\log P_y, \chi_{\ell m}) \ll x^{1-\varepsilon} (\log P_y)^2 \ll x^{1-\varepsilon} y^2.$$

Combining these with (5.5), we conclude

$$(5.9) \quad \sum_{\substack{x^{1/2} < p \leq x \log x \\ p \in \mathbb{P}_y}} 1 \geq \frac{x}{(\log x)2^{\pi(y)+2}} + O\left(x^{1-\varepsilon}2^{\pi(y)}y^2 + \frac{T_1(x, y) + T_4(x, y)}{(\log x)2^{\pi(y)}}\right)$$

uniformly for $x \geq 10$ and $1 \leq y \leq x^{1/2}$, where

$$\begin{aligned} T_\ell(x, y) &:= \sum_{m|P_y} \sum_{\substack{\rho(\ell m) \\ \beta(\ell m) \geq 1-\varepsilon, |\gamma(\ell m)| \leq \log P_y}} x^{\beta(\ell m)} \\ &= - \int_{1-\varepsilon}^1 x^\alpha dN_\ell(\alpha). \end{aligned}$$

It remains to estimate $T_\ell(x, y)$. From now on we take $y = y(x)$. By integration by parts and by using (5.8), we can deduce

$$(5.10) \quad T_\ell(x, y) \ll x^{1-\varepsilon}2^{\pi(y)}y^2 + x(\log x)I_\ell,$$

where

$$I_\ell := \int_0^\varepsilon x^{-\beta} N_\ell(1-\beta) d\beta.$$

Let $\eta = \eta(y)$ and $\eta_1 = \eta_1(y)$ be defined as in Lemmas 3 and 4, respectively. Set $\eta_2 := 2y(x)/(\log x) \log y$. It is easy to verify that $0 < \eta < \eta_1 < \eta_2 < \varepsilon$. (The inequality $\eta_1 < \eta_2$ governs the lower bound of $y(x)$ in (5.1).) Thus we can divide the interval $[0, \varepsilon]$ into four subintervals $[0, \eta]$, $[\eta, \eta_1]$, $[\eta_1, \eta_2]$ and $[\eta_2, \varepsilon]$, and denote by $I_{\ell,0}$, $I_{\ell,1}$, $I_{\ell,2}$ and $I_{\ell,3}$ the corresponding contribution to I_ℓ . Plainly we have

$$\frac{1}{2} \log_3 P_y \leq \frac{\eta}{4} \log x, \quad \frac{C_2 \log P_y}{\sqrt{\log_2 P_y}} \leq \frac{1}{4} \log x, \quad \frac{C_2 \log P_y}{\log(1/\eta_2)} \leq \frac{1}{2} \log x, \quad \frac{y}{\log y} = \frac{\eta_2}{2} \log x.$$

(The third inequality governs the upper bound of $y(x)$ in (5.1).) From Lemma 4.3 and (5.8), we deduce that

$$\begin{aligned} I_{\ell,1} &\ll \int_\eta^{\eta_1} \exp\left\{-\beta \log x + \frac{C_2 \beta \log P_y}{\sqrt{\log_2 P_y}} + \frac{1}{2} \log_3 P_y\right\} d\beta \ll \frac{x^{-\eta/2}}{\log x}, \\ I_{\ell,2} &\ll \int_{\eta_1}^{\eta_2} \exp\left\{-\beta \log x + \frac{C_2 \beta \log P_y}{\log(1/\beta)}\right\} d\beta \ll \frac{x^{-\eta_1/2}}{\log x}, \\ I_{\ell,3} &\ll \int_{\eta_2}^\varepsilon \exp\left\{-\beta \log x + \frac{y}{\log y}\right\} d\beta \ll \frac{x^{-\eta_2/2}}{\log x}. \end{aligned}$$

Hence, all of them satisfy

$$I_{\ell,i} = o((\log x)^{-1}) \quad (i = 1, 2, 3).$$

If we assume that both $\mathbf{L}_1(s, P_y)$ and $\mathbf{L}_4(s, P_y)$ have no zeros in the region (4.4) for all $y \geq 100$, then $I_{\ell,0} = 0$. Otherwise we use Lemma 4.2 to ensure the existence of $\{y_n\}_{n=1}^\infty$ such that $I_{\ell,0} = 0$.

With (5.10), our conclusion is

$$T_\ell(x_n, y_n) = o\left(\frac{x_n}{(\log x_n)2^{\pi(y_n)}}\right) \quad (n \rightarrow \infty),$$

or

$$T_\ell(x, y) = o\left(\frac{x}{(\log x)2^{\pi(y)}}\right) \quad (x \rightarrow \infty)$$

under the assumption that both $\mathbf{L}_1(s, P_y)$ and $\mathbf{L}_4(s, P_y)$ have no exceptional zeros. Clearly this and (5.9) imply the required result. This completes the proof of Proposition 5.1. $\square$

Now we are ready to prove Theorem 1.2.

Taking $Q_n = x_n \log x_n$ and $y(x) = 100\delta \log x$ in Proposition 5.1 and noticing that $p \in \mathbb{P}_y \Rightarrow n_{\chi_p} \geq y$, we have

$$\sum_{\substack{(Q_n/\log Q_n)^{1/2} < p \leq Q_n \\ n_{\chi_p} \geq 100\delta \log Q_n}} 1 \gg Q_n e^{-c_1(\log Q_n)/\log_2 Q_n}.$$

It implies the first assertion of Theorem 1.2, and the second one can be treated similarly. This concludes Theorem 1.2. $\square$

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DEPARTMENT OF MATHEMATICS, THE UNIVERSITY OF HONG KONG, POKFULAM ROAD, HONG KONG
E-mail address: yklau@maths.hku.hk

INSTITUT ELIE CARTAN, UNIVERSITÉ HENRI POINCARÉ (NANCY 1), 54506 VANDŒUVRE-LÈS-NANCY, FRANCE
E-mail address: wujie@iecn.u-nancy.fr